

Task 3: Basic SELECT Queries

K. Ameesha

Role: SQL Developer Intern

Task Overview & Metadata

Objective: Extract data from database tables using SELECT statements with filtering, sorting, and row limits.

Tools Used: DB Browser for SQLite / MySQL Workbench

Key Concepts: `SELECT` `WHERE` `ORDER BY` `LIMIT` `LIKE` `BETWEEN` `DISTINCT`

1. Key Concept Explanations

1. What does SELECT do?

The SELECT command fetches data records from a relational database table and returns specified attributes or columns in a tabular format.

2. How do you filter rows?

Rows are filtered using the WHERE clause by supplying logical comparison conditions (e.g., =, >, <, LIKE, IN, BETWEEN).

3. What is LIKE '%value%'?

It performs wildcard pattern matching to find records where the string contains "value" anywhere in the specified column (beginning, middle, or end).

4. What is BETWEEN used for?

BETWEEN is used to select values within a specified continuous range. It is inclusive of both the start and end boundary values.

5. How do you limit output rows?

By appending LIMIT n to the end of the query, restricting the maximum number of rows returned by the result set to n.

6. Difference between = and IN

The = operator checks against a single scalar value, while IN allows testing whether a value matches any element in a specified list.

7. How to sort in descending order?

Use the ORDER BY clause along with the DESC modifier (e.g., ORDER BY column_name DESC).

8. What is aliasing?

Aliasing provides a temporary alternative name for columns or tables using the AS keyword for better output readability and clarity.

9. Explain DISTINCT.

DISTINCT filters out duplicate rows from query results, ensuring that only unique values or combinations of values are displayed.

10. What is the default sort order?

The default sort order in SQL when using ORDER BY is **Ascending (ASC)**.

2. SQL Query Deliverable

```
-- =====
-- Elevate Labs: SQL Developer Internship
-- Task 3: Writing Basic SELECT Queries
-- Student Name: K. Ameesha
-- =====

-- 1. Retrieve all columns and specific column projections
SELECT * FROM products;

SELECT product_id, product_name, category, price, stock_quantity
FROM products;

-- 2. Data filtering using WHERE, AND, OR, LIKE, and BETWEEN
-- Filter using AND condition
SELECT * FROM products
WHERE category = 'Electronics' AND price > 15000;

-- Filter using OR condition
SELECT * FROM products
WHERE category = 'Electronics' OR category = 'Appliances';

-- Substring search using LIKE
SELECT * FROM products
WHERE product_name LIKE '%Smart%';

-- Range filtering using BETWEEN
```

```
SELECT * FROM products
WHERE price BETWEEN 5000 AND 25000;

-- 3. Sorting and Limiting Output
SELECT product_id, product_name, price
FROM products
WHERE stock_quantity > 0
ORDER BY price DESC
LIMIT 5;

-- 4. Using DISTINCT for unique values and Column Aliasing
SELECT DISTINCT category AS product_category
FROM products;
```